

* Secant Method : (Chord Method)

Ex. A real root of equation $f(x) = x^3 + 5x + 1 = 0$ lies in $(-1, 0)$. Perform five iterations of the secant method to obtain its root.

→ let,

$$f(x) = x^3 + 5x + 1 = 0$$

$$x \quad -2 \quad -1 \quad 0 \quad 1 \quad 2$$

$$f(x) \quad -17 \quad -5 \quad 1 \quad 7 \quad 19$$

Now, The root of $f(x) = 0$ lies in $(-1, 0)$
we have,

$$x_0 = -1, \quad x_1 = 0$$

$$f(x_0) = f(-1) = -5 \quad f(x_1) = f(0) = 1$$

we know that iteration formula of the secant method.

$$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}, \quad n = 1, 2, \dots$$

1st iteration:

$$x_2 = x_1 - \frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$= 0 - \frac{(1)(0 - (-1))}{1 - (-5)}$$

$$= -\frac{1}{6}$$

$$x_2 = -\frac{1}{6} = -0.1667$$

$$f(x_2) = 0.1618$$

2nd iteration:

$$x_3 = x_2 - \frac{f(x_2)(x_2 - x_1)}{f(x_2) - f(x_1)}$$

$$= -0.1667 - \frac{(0.1618)(-0.1667 - 0)}{0.1618 - 1} = -0.1989$$

$$f(x_3) = -0.0024$$

3rd iteration:

$$x_4 = x_3 - \frac{f(x_3)(x_3 - x_2)}{f(x_3) - f(x_2)}$$

$$= -0.1989 - \frac{(-0.0024)(-0.1989 - (-0.1667))}{(-0.0024 - (-0.1667))}$$

$$x_4 = -0.1985$$

$$f(x_4) = -0.00032$$

4th iteration:

$$x_5 = x_4 - \frac{f(x_4)(x_4 - x_3)}{f(x_4) - f(x_3)}$$

$$= -0.1985 - \frac{(-0.00032)(-0.1985 - (-0.1989))}{(-0.00032 - (-0.0024))}$$

$$x_5 = -0.1983$$

$$f(x_5) = 0.0007$$

5th iteration:

$$x_6 = x_5 - \frac{f(x_5)(x_5 - x_4)}{f(x_5) - f(x_4)}$$

$$f(x_5) - f(x_4)$$

$$= -0.1983 - \frac{(0.0007)(-0.1983 - (-0.1985))}{(0.0007 - (-0.00032))}$$

$$x_6 = -0.1982$$

$$f(x_6) = 0.0012$$